Modeling Wildfire Spread Using Asymmetric Cell-DEVS

Manthan Patel
School of Information Technology
Carleton University, Ottawa, Canada
GitHub: https://github.com/PatelManthan98/Wildfire_Asym_Cell_Devs.git

***Abstract*—Wildfire activity is intensifying due to climatic extremes, fuel accumulation, and complex terrain, producing extreme events whose directional, multi-scale behaviour routine cellular models struggle to reproduce. This study presents a physics-informed wildfire spread model based on the Asymmetric Cell-DEVS formalism and implemented in the Cadmium v2 simulator. Directional processes—wind forcing, slope-induced acceleration, and ember transport—are encoded as per-edge vicinity weights in terrain-derived asymmetric topologies generated from NRCan CDEM and NLCMS-2015 land-cover data; the model also incorporates a five-class fuel scheme with fuel-dependent burn durations and a probabilistic ember-spotting mechanism with distance-decaying, wind-aligned edges. We evaluate the model across seven scenarios on a 50×50 grid at $100\,\mathrm{m}$ resolution, spanning calm baselines, wind-driven grass fires, river firebreaks with and without spotting, wildland–urban interface (WUI) configurations, and a reconstruction of the 2016 Fort McMurray Horse River wildfire. Results show that embedding directional physics in the topology produces isotropic spread under calm conditions, sharply anisotropic plumes under wind, structural containment by rivers, threshold-driven spotting breaches, and rapid WUI escalation, while reproducing the spotting-driven river crossing that characterised the Fort McMurray event. The asymmetric formulation thus offers a credible and computationally efficient platform for scenario analysis, mitigation planning, and pedagogical exploration of wildfire dynamics.**

***Index Terms*—Wildfire modelling, Asymmetric Cell-DEVS, Rothermel model, ember spotting, terrain-derived topology, Cadmium v2, NRCan DEM, NLCMS-2015.**

I. Introduction

Wildfires represent one of the most complex and destructive natural hazards affecting ecosystems, infrastructure, and communities across Canada and globally. Their behaviour emerges from nonlinear interactions among fuel, weather, and terrain, producing highly dynamic and often unpredictable spread patterns. In recent decades, wildfire activity has intensified due to climate-driven increases in temperature, prolonged droughts, and fuel accumulation. These trends have contributed to a growing number of extreme events characterized by rapid fire growth, long-range ember spotting, and large-scale evacuations.

The 2016 Fort McMurray Horse River wildfire remains one of the most significant Canadian examples of such behaviour, resulting in unprecedented economic losses, widespread structural damage, and the displacement of more than 80,000 residents [11]. Post-event analyses emphasised that the fire's most consequential behaviour—its crossing of the wide Athabasca River and subsequent ignition of residential neighbourhoods—was driven almost entirely by ember transport, not by contiguous flame spread. Similar spotting-dominated behaviour has been documented in the 2003 Canberra firestorm, the 2018 Camp Fire in California, and the 2021 Lytton event in British Columbia. These events highlight the urgent need for modelling frameworks capable of capturing the directional, multi-scale, and stochastic processes that govern wildfire spread.

From a simulation perspective, the challenge is that the processes which matter most during extreme events operate at fundamentally different scales. Flame-front propagation is a local, near-neighbour phenomenon; wind-driven spread introduces strong directional bias at the intermediate scale; and firebrand transport produces sparse, non-local connections that can leapfrog natural and engineered barriers. Any single-scale or purely local formalism will struggle to reproduce the qualitative character of such events.

A. Motivation

Traditional wildfire spread models range from empirical rate-of-spread formulations to physics-based computational fluid dynamics (CFD) simulations. Empirical models such as Rothermel's equation [13] provide operationally useful estimates of surface fire behaviour but lack spatial explicitness and cannot directly represent complex terrain or heterogeneous fuels. At the opposite end of the spectrum, CFD-based models such as FIRETEC and WFDS offer detailed representations of plume dynamics and heat transfer [14] but are computationally prohibitive for large-scale or real-time applications. Operational tools such as FARSITE [6] occupy an intermediate niche, discretising the landscape into polygonal fronts that advance according to Rothermel-derived rates, but they still lack the explicit asynchronous timing and directional topology needed to study extreme, spotting-dominated behaviour.

Cellular automata (CA) and Cell-DEVS models offer a middle ground by discretizing the landscape into interacting cells with local transition rules [1], [15]. However, most CA-based wildfire models assume symmetric neighbourhoods, where the influence of cell A on cell B equals the influence of B on A. This assumption is fundamentally incompatible with wildfire physics: wind, slope, and ember transport introduce strong directional biases that cannot be captured by symmetric influence structures. As a result, symmetric CA models often produce unrealistic, near-circular spread patterns even under strong wind forcing, fail to reproduce wind-aligned fire runs, and cannot represent barrier crossings driven by spotting.

The motivation for this work is to develop a wildfire modelling framework that preserves the computational efficiency of cellular models while incorporating the directional, physics-driven processes that dominate real wildfire behaviour. Asymmetric Cell-DEVS provides a natural foundation for this objective because it allows per-edge weights and one-way influence to be baked into the topology itself, rather than encoded as special cases inside transition functions.

B. Why Asymmetric Cell-DEVS

Asymmetric Cell-DEVS extends the classical Cell-DEVS formalism [2], [3] by allowing each cell to define a unique, directed neighbourhood with per-edge vicinity weights. These weights encode the strength and direction of influence from neighbouring cells, enabling the model to structurally represent wind forcing, slope-induced acceleration, and long-range ember spotting. Unlike symmetric CA models, where directional effects must be approximated through conditional logic inside transition functions, Asymmetric Cell-DEVS embeds these effects directly into the topology. This approach offers several advantages:

- **Directional realism:** Wind and slope effects are represented as asymmetric weights, allowing downwind and uphill spread to dominate naturally without if-else branching.
- **Structural firebreaks:** Rivers, roads, and non-flammable surfaces are encoded as zero-weight edges, eliminating the need for repeated conditional checks inside the transition function.
- **Long-range interactions:** Ember spotting is represented as sparse, directed edges that connect burning cells to downwind targets beyond the Moore neighbourhood, capturing jump-type behaviour that purely local rules cannot express.
- **Reduced transition complexity:** Embedding physics in the topology simplifies the transition function and improves computational efficiency because ignition probability becomes a small dot-product-like computation over pre-weighted edges.
- **Separation of concerns:** Terrain, fuel, and meteorology are compiled into the topology once; the simulation loop then only handles state transitions, which makes reproducibility and sensitivity analysis straightforward.

These properties make Asymmetric Cell-DEVS particularly well suited for modelling heterogeneous landscapes where directional processes dominate fire behaviour.

C. Research Gap

Despite the advantages of cellular and DEVS-based wildfire models, several gaps remain in the literature:

- Many existing CA and Cell-DEVS models rely on symmetric neighbourhoods, limiting their ability to represent directional spread [4], [5].
- Spotting is often ignored or treated as a simple probabilistic jump, without incorporating wind alignment, fuel type, or distance decay [8], [9].
- Few models integrate real terrain and land-cover data into the topology itself, reducing their applicability to real-world scenarios and requiring separate preprocessing pipelines.
- Validation against documented extreme events—particularly those involving barrier crossings—is limited, so the qualitative regime transitions characteristic of catastrophic fires are rarely reproduced.

This work addresses these gaps by combining terrain-derived asymmetric topologies, Rothermel-based directional physics, and a probabilistic spotting mechanism within a unified Asymmetric Cell-DEVS framework.

D. Goals and Contributions

The primary goal of this research is to develop a physics-informed, terrain-aware wildfire spread model using the Asymmetric Cell-DEVS formalism and to evaluate its ability to reproduce realistic fire behaviour across a range of scenarios. The key contributions of this work are:

- **A terrain processing pipeline** that converts NRCan CDEM elevation data and NLCMS-2015 land-cover rasters into asymmetric JSON topologies with per-edge vicinity weights encoding wind alignment, slope angle, and distance.
- **A five-category fuel classification** mapped from land-cover data, enabling differentiated ignition probability, burn duration, and ember production across fuel types.
- **Integration of Rothermel wind and slope factors** into per-pair vicinity weights, embedding directional physics structurally rather than procedurally.
- **A probabilistic ember-spotting mechanism** that generates sparse, long-range downwind edges with distance decay and fuel-dependent spotting ability.
- **A complete Cadmium v2 implementation** with inertial delays, reproducible scenario JSONs, and a Python visualization suite for generating GIFs, PNGs, and temporal metrics.
- **Validation through seven scenarios**, including a reconstruction of the 2016 Fort McMurray wildfire, demonstrating the model's ability to reproduce

spotting-driven river crossings and the qualitative regime change between contained and catastrophic outcomes.

E. Paper Organization

The remainder of this paper is organized as follows. Section II reviews wildfire physics, the Rothermel model, the FWI system, and the Cell-DEVS formalism. Section III presents the conceptual and formal Asymmetric Cell-DEVS wildfire model, including vicinity-weight derivation, transition rules, and the Cadmium v2 implementation. Section IV presents simulation scenarios and results, including a Fort McMurray validation study and a sensitivity analysis. Section V interprets the results, Section VI outlines limitations, Section VII sketches future work, and Section VIII concludes.

II. Background

Wildfire spread is governed by a complex interplay of physical, ecological, and atmospheric processes that operate across multiple spatial and temporal scales. Modelling these processes requires a balance between physical realism, computational efficiency, and the ability to incorporate heterogeneous terrain and fuel conditions. This section reviews the fundamental drivers of wildfire behaviour, the semi-empirical and cellular modelling approaches commonly used in fire science, and the theoretical foundations of the Asymmetric Cell-DEVS formalism adopted in this work.

A. Wildfire Physics and Rate-of-Spread Fundamentals

Wildfire behaviour emerges from coupled interactions among three primary components—the *fire behaviour triangle* of *fuel*, *weather*, and *topography*. Fuel characteristics (load, continuity, arrangement, size class, and moisture content) determine the amount of energy available for combustion and the ease with which fire can propagate [7]. Weather variables (wind speed, wind direction, temperature, and relative humidity) influence flame tilt, convective heat transfer, and fuel drying. Topography modifies local wind fields and affects preheating of fuels, with uphill slopes significantly accelerating fire spread.

At the cell scale, three heat-transfer modes contribute to ignition of an unburned neighbour: *radiation* from the flame front, *convection* of hot gases tilted by wind and slope, and *firebrand transport* lofted by the convective plume. Radiation dominates at short range and is relatively isotropic; convection is strongly directional and increases sharply with wind speed; firebrand transport is stochastic and can reach distances far beyond any continuous flame. A faithful spread model must therefore blend short-range isotropic influence, short-range anisotropic influence, and sparse long-range influence. The combined effect of these factors determines the *rate of spread* (ROS), fireline intensity, and the likelihood of extreme behaviours such as crown fire, vortex formation, and ember lofting.

Operational fire behaviour models typically rely on semi-empirical formulations that relate ROS to measurable environmental inputs [6]. In spatially explicit simulations, ROS is often converted into probabilistic ignition rules or vicinity weights that determine the likelihood that a burning cell will ignite a neighbour during a time step—a conversion that is particularly natural in a discrete-event cellular formalism.

B. Rothermel Wind and Slope Enhancement Factors

Rothermel's surface fire spread model [13] remains one of the most widely used semi-empirical formulations in wildfire science. The model expresses ROS as

$$R = \frac{I_R\,\xi\,(1+\phi_w+\phi_s)}{\rho_b\,\varepsilon\,Q_{ig}}, \tag{1}$$

where I_R is the reaction intensity, ξ the propagating flux ratio, ρ_b the bulk density, ε the effective heating number, Q_{ig} the heat of pre-ignition, and ϕ_w, ϕ_s are the wind and slope enhancement factors respectively. The wind factor ϕ_w increases ROS when wind aligns with the direction of spread and decreases ROS (or contributes zero) when wind opposes it. The slope factor ϕ_s is typically expressed as a function of the square of the tangent of the slope angle, producing strong uphill enhancement and weaker downhill suppression.

In cellular models, these enhancement factors can be embedded directly into per-pair vicinity weights. Rather than recomputing ϕ_w and ϕ_s inside every transition call, the model compiles a directional weight $w_{n\to c}$ once during topology generation. This structural embedding is particularly important for representing wind-driven fire runs and slope-accelerated spread in heterogeneous terrain, because it decouples expensive geometric calculations from the hot simulation loop.

C. Canadian Forest Fire Weather Index (FWI) System

The Canadian Forest Fire Weather Index (FWI) System provides standardized indices that quantify fuel moisture and fire danger based on routine meteorological observations. The FWI system comprises six components arranged in a cascade: the Fine Fuel Moisture Code (FFMC), the Duff Moisture Code (DMC), the Drought Code (DC), the Initial Spread Index (ISI), the Buildup Index (BUI), and the overall FWI.

For short-term spread modelling, the FFMC is especially relevant to ignition probability because it represents the moisture content of fine surface fuels that ignite readily. FFMC values near 80 indicate damp fuels that require substantial preheating; values above 90 indicate critically dry fuels; and values above 95 correspond to fuels in which every spark has a high probability of sustained ignition. High FFMC values are a near-universal precondition for extreme events: the Fort McMurray fire occurred at FFMC ≈ 97. Integrating FFMC (or a normalized variant) into ignition probability formulations ensures that the model responds realistically to weather-driven changes in flammability.

D. Challenges in Modelling Wildfire Spread

Despite decades of research, modelling wildfire spread remains challenging due to:

- **Strong directionality:** Wind and slope create highly anisotropic spread patterns that symmetric neighbourhoods cannot represent.
- **Multi-scale interactions:** Spotting introduces long-range, non-local ignitions that violate the assumptions of local neighbourhood models.
- **Heterogeneous landscapes:** Real terrain includes rivers, roads, wetlands, and urban structures that act as barriers or accelerants [12].
- **Stochastic behaviour:** Ignition, ember lofting, and firebrand landing are inherently probabilistic, so deterministic rules tend to produce overly smooth fronts.
- **Threshold regime changes:** Many real fires exhibit qualitative jumps (e.g., crossing a river) that depend sensitively on a few parameters, making smooth interpolation of outputs misleading.

These challenges motivate the use of modelling frameworks that support directional influence, heterogeneous neighbourhoods, and stochastic long-range interactions.

E. Cell-DEVS Formalism

Cell-DEVS [3] extends classical cellular automata by embedding each cell as an atomic DEVS model with explicit timing, internal and external transitions, and output functions. Unlike synchronous CA, Cell-DEVS supports:

- **Asynchronous updates** through time-advance functions, so idle areas consume no compute budget.
- **Inertial and transport delays** that model combustion duration and heat transfer.
- **Complex state variables** such as intensity, moisture, and burn duration.
- **Hierarchical composition** for multi-scale modelling.

These features make Cell-DEVS well suited for processes with heterogeneous temporal scales, such as short-duration grass fires versus long-duration forest burns, and for mixing fine-grained cell dynamics with coarse-grained weather or suppression models.

F. Limitations of Symmetric Cellular Models

Traditional CA and symmetric Cell-DEVS models assume reciprocal influence between neighbouring cells. This assumption is problematic for wildfire modelling because:

- Wind and slope effects are inherently directional and routinely produce 5:1 or higher ratios between downwind and upwind propagation rates.
- Ember spotting produces long-range, one-way influence: a burning forest can ignite a neighbourhood one kilometre downwind, but the reverse is not true.
- Firebreaks require zero-weight edges, not symmetric neighbourhoods, because a river that stops ground-borne spread does nothing to stop firebrands.
- Symmetric influence often produces unrealistic circular spread patterns even under strong wind forcing.

These limitations motivate the use of asymmetric neighbourhoods.

G. Asymmetric Cell-DEVS and Directional Topologies

Asymmetric Cell-DEVS generalizes the Cell-DEVS formalism by allowing each cell to define a unique, directed neighbourhood with per-edge vicinity weights. These weights encode the strength and direction of influence from neighbouring cells. This approach enables:

- **Wind-aligned spread** through directional amplification of downwind edges.
- **Slope-driven acceleration** through elevation-derived weights that favour upslope transfer.
- **Structural firebreaks** via zero-weight edges across rivers or roads.
- **Long-range spotting** through sparse, directed edges beyond the Moore neighbourhood.

Embedding directional physics in the topology simplifies transition functions and improves computational efficiency. It also makes the model easier to audit: a domain expert can inspect the compiled JSON and verify that the river's edges are indeed zero, that the spotting edges point in the correct direction, and that the vicinity weights align with the prevailing wind.

H. Ember Spotting: Mechanisms and Modelling Approaches

Ember spotting occurs when burning firebrands are lofted into the convective plume and transported downwind [8]. Spotting distances depend on:

- Fire intensity and plume strength,
- Ember size, mass, and aerodynamic properties,
- Wind speed and direction,
- Fuel receptivity at the landing site.

Modelling approaches include:

- **Stochastic kernels:** Probabilistic distributions for ember landing distance.
- **Plume-coupled models:** CFD-based simulations of ember trajectories [10].
- **Directed spot edges:** Sparse, long-range edges with distance decay (used in this work).

The directed-edge approach balances realism and computational efficiency, enabling barrier crossings without introducing spurious ignitions. By capping the maximum spotting range and applying a $1/d^{1.5}$ decay, the model produces a realistic distribution of spot ignitions concentrated near the flame front and dwindling at longer distances.

I. Comparison of Modelling Paradigms

Table I summarizes key differences among CA, symmetric Cell-DEVS, and Asymmetric Cell-DEVS.

TABLE I: Comparison of wildfire modelling paradigms.

Model Type	Strengths	Limitations
CA	Simple, fast	No timing; symmetric; limited physics
Symmetric Cell-DEVS	Explicit timing and delays	Cannot represent directional spread
Asymmetric Cell-DEVS	Directional physics, spotting, firebreaks	Requires topology generation

J. Related Work

Previous studies have integrated GIS-derived terrain and fuel data into cellular and DEVS-based wildfire models [1], [6]. Research themes include embedding Rothermel-based physics into spatial models, using asymmetric neighbourhoods to represent directional processes, and modelling spotting through empirical or plume-coupled approaches [9], [10]. Recent work has highlighted the importance of spotting in river-crossing events and recommended explicit spotting mechanisms or higher-resolution plume models [11]. This project synthesizes these advances by combining terrain-derived asymmetric topologies, Rothermel-based directional physics, and a parameterized spotting mechanism within an Asymmetric Cell-DEVS implementation, with an explicit focus on reproducing the qualitative regime change between contained and catastrophic outcomes.

III. Models Defined

Wildfire spread emerges from interacting physical processes that operate across multiple spatial and temporal scales. The Asymmetric Cell-DEVS formalism provides a natural foundation for representing these processes by embedding directional influence, heterogeneous neighbourhoods, and explicit timing into a spatially distributed model. This section presents the conceptual model, the formal DEVS specification, the state variables and parameters, the transition rules, and the algorithmic flow governing the wildfire simulation.

A. Conceptual Model of Wildfire Spread

The conceptual model integrates environmental, fuel-based, and physical fire-behaviour processes to represent how wildfires ignite, intensify, and propagate across heterogeneous terrain. The landscape is discretized into a grid of $100\,\mathrm{m} \times 100\,\mathrm{m}$ cells, each representing a spatial unit with static environmental attributes (fuel type, elevation, moisture) and dynamic fire-behaviour states (unburned, burning, or ash).

Fuel classifications derived from NLCMS-2015 land-cover data determine the flammability, burn duration, and ember-production potential of each cell. Elevation values from NRCan CDEM are used to compute slope and aspect, which influence directional fire spread. Weather variables such as wind speed, wind direction, temperature, relative humidity, and FFMC modulate ignition probability and fire intensity.

A defining characteristic of wildfire behaviour is its *directionality*. Wind tilts flames toward downwind fuels, increasing convective and radiant heat transfer. Slope accelerates uphill spread by preheating fuels above the flame front. Ember spotting introduces long-range, non-local ignitions that bypass natural firebreaks. These directional processes are embedded structurally in the model through asymmetric neighbourhood weights, allowing the influence of cell A on cell B to differ from the influence of B on A.

These interacting processes—fuel dynamics, wind-driven spread, slope effects, ember spotting, and environmental modulation—form the foundation for the formal Asymmetric Cell-DEVS specification.

B. Formal Specification of the Asymmetric Cell-DEVS Model

Each cell is represented as an atomic DEVS model:

$$WildfireCell = \langle X,\, Y,\, S,\, N,\, delay,\, \delta_{int},\, \delta_{ext},\, \delta_{con},\, \lambda,\, D \rangle \tag{2}$$

where:

- X: external inputs (ignition events, neighbour updates),
- Y: outputs (state changes broadcast to neighbours),
- S: state vector containing static and dynamic variables,
- N: asymmetric neighbourhood with per-edge vicinity weights,
- $delay$: inertial delays for combustion,
- $\delta_{int}, \delta_{ext}, \delta_{con}$: internal, external, and confluent transitions,
- λ: output function,
- D: time-advance function.

The coupled model for the entire grid is:

$$WildfireModel = \langle X_{list},\, Y_{list},\, I,\, X,\, Y,\, \eta,\, \{t_1, \ldots, t_n\},\, N,\, C,\, B,\, Z \rangle \tag{3}$$

The translation function Z computes directional vicinity weights using wind alignment, slope angle, distance, and fuel type, and is applied once at topology-construction time. The asymmetric neighbourhood N is therefore a directed graph whose edges are pre-weighted; the runtime model treats these weights as constants until they are re-compiled (e.g., for a new wind regime).

C. State Variables and Parameters

Each cell maintains a set of static and dynamic variables:

- **Fire state** $state \in \{0, 1, 2, 3\}$: nonflammable, unburned, burning, ash.
- **Intensity** $I \in [0, 1]$: relative combustion strength, used to modulate the cell's outgoing ignition probability.
- **Burn duration** b: countdown of burn steps based on fuel type.

TABLE II: Fuel-dependent parameters used in this model.

Fuel Type	Burn Steps	Intensity	Spot Ability
Grass	3	0.4	Low
Shrub	5	0.6	Medium
Forest	8	0.8	High
Urban	10	1.0	Very High
Water/Road	0	0.0	None

- **Fuel type** f: one of five categories derived from NLCMS-2015.
- **Elevation** z: used to compute slope and aspect.
- **Fuel moisture** m: influences ignition probability and is initialized from FWI-derived proxies.
- **Weather parameters**: FFMC, temperature T, humidity H, wind speed W_s, wind direction W_d.
- **Spotting parameters**: $spot_base$, $spot_range$, $spot_ability(f)$.

The fuel parameters in Table II are consistent with the qualitative ordering in standard fuel model families: grass fires are short-lived and low-intensity but spread rapidly, forest crowns burn long and intensely and produce many embers, and urban cells are treated as worst-case because structural fuel loads can sustain near-continuous burning for long periods.

D. Vicinity Weight Computation

Directional vicinity weights represent the degree of influence a burning neighbour has on its surrounding cells. Rather than calculating complex physics during every second of the simulation, these weights are precomputed to represent the structural "ease" with which fire moves in a specific direction based on four primary factors:

Distance Factor: The model accounts for the physical distance between cell centres. In a standard eight-neighbour (Moore) grid, cells are not equidistant; orthogonal neighbours (up, down, left, right) are closer than those located diagonally. The weight is inversely proportional to distance, ensuring that fire naturally prefers to spread to its closest immediate neighbours before reaching out to diagonal ones.

Wind Alignment Factor: Wind is a dominant driver of fire directionality. This factor calculates how closely the direction of a specific neighbour aligns with the current wind vector. If a neighbour is directly downwind, the weight is significantly amplified to simulate convective heat transfer. Conversely, if the wind is blowing away from a neighbour, the wind contribution is removed entirely, meaning spread in that direction must rely solely on other factors like slope or fuel density.

Slope Factor: To simulate the physics of fire moving uphill, the model incorporates a slope enhancement factor based on terrain elevation. Fire typically spreads much faster uphill because the flames are tilted closer to the unburned fuel, preheating it more effectively. This factor produces a strong weight increase for uphill neighbours while maintaining a baseline value for spread on flat ground or downhill.

Fuel Modifier: The final weight component is determined by the specific land cover of the target cell. This allows the model to structurally define barriers and accelerators. For example, the weight is set to zero for non-flammable surfaces like water or wide roads, effectively breaking the connection in the graph. In contrast, high-density fuels like forests or urban structures receive a larger modifier to represent their high fuel load and potential for intense spread.

By precomputing these combined factors into a single constant weight for every connection in the grid, the C++ simulation engine can process complex, asymmetrical fire spread with extreme efficiency, requiring only basic arithmetic during the active simulation phase.

E. Transition Rules and Fire Spread Dynamics

1) Combustion Dynamics: This section defines the lifecycle of a cell after it has ignited. Rather than burning out instantly, the model simulates a realistic progression of fire. Once a cell catches fire, it begins a countdown based on its specific fuel type. During this time, the fire's intensity follows a natural curve: it builds up from ignition, reaches a peak plateau of maximum heat, and then slowly decays until the fuel is exhausted.

2) Neighbour-Driven Ignition: The core of the local fire spread is governed by a factored probability model that determines if a burning cell will ignite its immediate neighbours. This model balances atmospheric conditions (FFMC, temperature, and humidity), the specific fuel moisture on the receiving side, and the directional physics defined in the topology.

3) Ember Spotting: To account for long-distance fire spread that bypasses traditional barriers like roads or rivers, the model includes an ember spotting mechanism. This is represented by special directed connections in the topology that follow the wind vector. Embers are concentrated in a downwind cone; while most land nearby, an empirical decay function allows for rare, long-distance jumps.

4) Environmental Modulation: As a final safeguard, a global modulation factor acts as a "safety net." This ensures that the combined mathematical corrections do not produce unrealistic results during extreme weather events, dampening probabilities in cold/wet conditions and amplifying them during extreme heat to represent nonlinear "blow-up" behavior.

F. Algorithmic Flow of the Model

Algorithm 1 summarises the per-step update for a single cell. The outer conditional is a cheap dispatch on fire state;

the inner loops iterate only over the cell's actual neighbours, ensuring performance scales with the topology's complexity rather than just the grid size.

Algorithm 1 Wildfire Cell Update

```
Input: state S, neighbours N, weather W, fuel f
if state = burning then
    decrement burn duration
    if b = 0 then
        state ← ash
    end if
else
    for each neighbour n ∈ N do
        compute P_{n→c}
        if RAND() < P_{n→c} then
            state ← burning
        end if
    end for
    for each spotting edge n do
        compute w^{spot}_{n→c}
        if RAND() < w^{spot}_{n→c} then
            state ← burning
        end if
    end for
end if
```

G. Implementation in Cadmium v2

The wildfire model is implemented in Cadmium v2, a C++17 discrete-event simulation engine that supports hierarchical DEVS and Cell-DEVS models with efficient event scheduling. The implementation consists of four major components: (1) the atomic wildfire cell class, (2) the coupled grid model, (3) the topology and terrain loader, and (4) the simulation runner and logging subsystem.

Atomic Cell Implementation: Each grid cell is implemented as a C++ atomic model derived from `cadmium::cell_atomic`. The class encapsulates the full state vector described above, including fire state, intensity, burn duration, fuel type, moisture, and static terrain attributes. The internal transition function (`delta_int`) decrements burn duration and handles the transition from burning to ash. The external transition function (`delta_ext`) processes ignition attempts from neighbours and spotting edges. The confluent transition function (`delta_con`) resolves simultaneous internal and external events by applying the internal transition first and then the external one, which ensures that a cell that is scheduled to become ash in the same instant that it receives an ignition message does not spuriously re-ignite. The output function (`lambda`) emits state-change messages that propagate to neighbouring cells.

Cadmium's event-driven architecture ensures that only cells experiencing state changes are scheduled for updates, significantly reducing computational overhead compared to synchronous cellular automata. In practice, once the active fire front stabilises, the number of events per step is proportional to the perimeter of the front rather than the area of the grid.

Coupled Grid Model: The grid is implemented as a coupled DEVS model composed of N atomic wildfire cells arranged in a 2D layout. The coupled model defines the message-routing structure between cells using the asymmetric neighbourhood specified in the topology JSON. Each directed edge in the topology corresponds to a Cadmium port-to-port coupling, enabling directional influence without requiring conditional logic inside the atomic model.

This design allows the grid to scale to thousands of cells while maintaining efficient event propagation. Firebreaks, rivers, and nonflammable surfaces are represented structurally as missing or zero-weight edges, requiring no special-case logic.

Topology and Terrain Loader: A Python preprocessing pipeline generates a JSON file containing:

- the asymmetric neighbourhood for each cell,
- fuel type and moisture values,
- elevation, slope, and aspect,
- precomputed vicinity weights ($w_{n\to c}$),
- spotting edges and spotting parameters.

The C++ loader parses this JSON at runtime and constructs the grid model accordingly. All static attributes are stored directly in each cell's state, while dynamic attributes (e.g., burn duration, intensity) are initialized based on scenario parameters. The JSON acts as a single source of truth: changing terrain, land cover, or wind regime requires re-running the pipeline, but not recompiling the C++ code.

Simulation Runner and Logging: Simulations are executed using Cadmium's `root_coordinator`, which manages global time advancement and event scheduling. The runner accepts command-line arguments specifying the simulation duration, random seed, ignition points, weather parameters, and topology file path. Cadmium's built-in logging system records state transitions, ignition events, timestamps of internal and external transitions, and per-cell outputs. Logs are written to CSV files and processed by a Python visualization suite that generates per-step PNG frames, animated GIFs of fire spread, temporal plots of active burning cells, and cumulative burned-area curves.

Reproducibility and Parameterization: All simulations are fully reproducible due to explicit seeding of Cadmium's random number generator. Scenario parameters (wind, humidity, FFMC, spotting strength, ignition location) are stored in scenario JSON files, enabling batch experiments and comparative studies. The modular design allows new fuel types, spotting kernels, or ignition rules to be added with minimal changes to the core model. Overall, the Cadmium v2 implementation provides a flexible, efficient, and extensible platform for simulating wildfire spread under heterogeneous terrain and extreme environmental conditions.

TABLE III: Quantitative results for all scenarios (seed = 42, sim_time = 500). Scenario 7 burned 222 urban cells.

Scenario	Burned	% Grid	Peak Fire	Pattern
1. Calm Forest	2447	97.9%	377	Radial symmetric
2. Windy Grassland	2323	92.9%	228	E corridor
3. Firebreaks	1124	45.0%	187	Contained W
4. Firebreaks + Spot	1143	45.7%	195	Partial breach
5. Urban Interface	2383	95.3%	254	WUI breach
6. FtMcM no spot	807	32.3%	264	River contains
7. FtMcM + spot	1791	71.6%	334	River crossed

IV. Simulation Results

A. Experimental Design and Scenarios

Seven scenarios were designed to evaluate the behaviour of the Asymmetric Cell-DEVS wildfire model under different environmental and structural conditions. Each scenario was run for 500 simulation steps on a 50×50 grid (100 m resolution), using `seed = 42` for reproducibility. The scenarios isolate the effects of wind, fuel type, firebreaks, spotting, and urban structures, allowing a controlled comparison of how each factor influences fire dynamics. Scenarios 1 through 5 are synthetic configurations designed to probe individual mechanisms, while Scenarios 6 and 7 reconstruct the geography and meteorology of the 2016 Fort McMurray wildfire.

B. Quantitative Overview

Table III summarizes the quantitative outcomes for all seven scenarios. Burned-area fractions range from 32.3% (Fort McMurray without spotting) to 97.9% (calm forest), illustrating that the same cellular substrate can produce radically different outcomes depending on which physical mechanisms the topology encodes.

C. Scenario 1 — Calm Forest (Symmetric Baseline)

This scenario establishes the baseline behaviour of the model under symmetric conditions. With no wind and a homogeneous forest fuel class, vicinity weights are effectively isotropic and the ignition probability modifiers are neutral. As shown in Fig. 1, the fire expands outward in a near-circular front from the ignition cell.

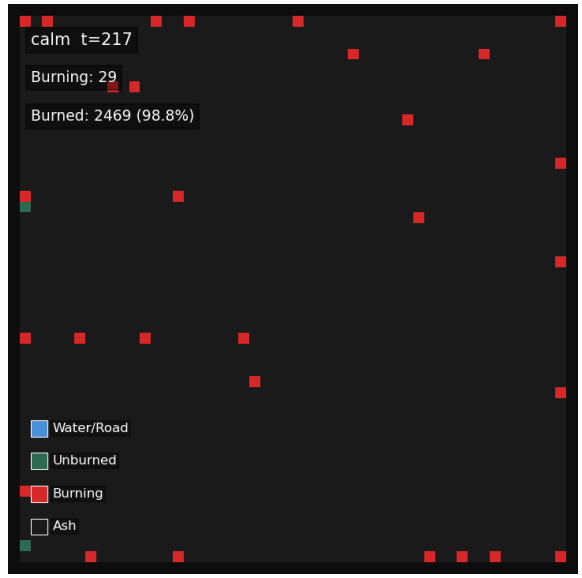

Fig. 1: Scenario 1 (Calm Forest): symmetric radial spread under no-wind conditions.

D. Scenario 2 — Windy Grassland

This scenario imposes a 35 km/h east wind over a grass-dominated landscape. Grass fuels have short burn durations and low fuel loads, making them highly responsive to wind-driven convective and radiative transfer. The fire does not wrap back on itself or exhibit any upwind growth.

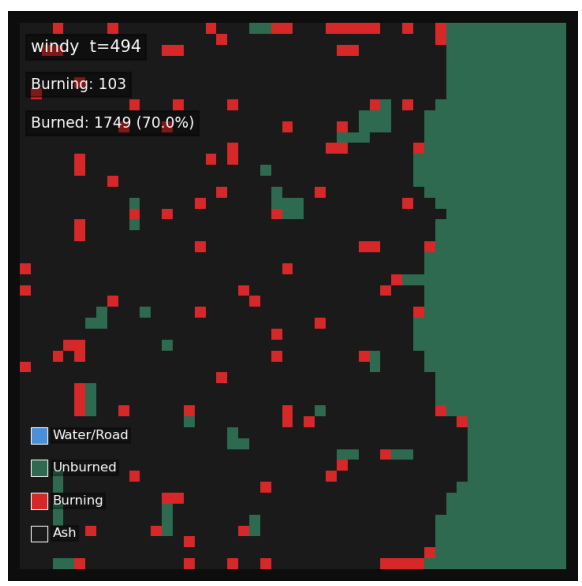

Fig. 2: Scenario 2 (Windy Grassland): eastward plume driven by 35 km/h wind; patchy lateral spread due to rapid grass combustion.

E. Scenario 3 — Firebreaks (No Spotting)

This scenario introduces a river as a structural firebreak (zero-vicinity edges). The fire advances until it reaches the river and then halts cleanly along the bank as shown in Fig. 3.

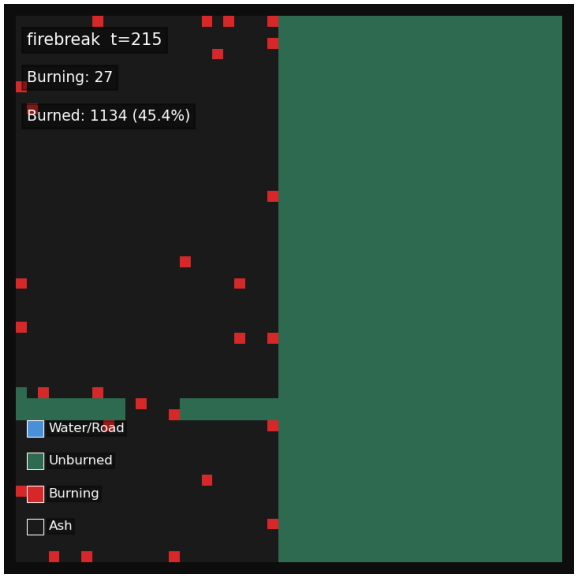

Fig. 3: Scenario 3 (Firebreaks): river firebreak fully contains the fire; containment emerges from asymmetric topology.

F. Scenario 4 — Firebreaks + Spotting

Spotting is enabled with moderate wind (25 km/h). Occasional embers cross the river but fail to sustain spread, because the sparse spot ignitions extinguish before they can connect into a contiguous front.

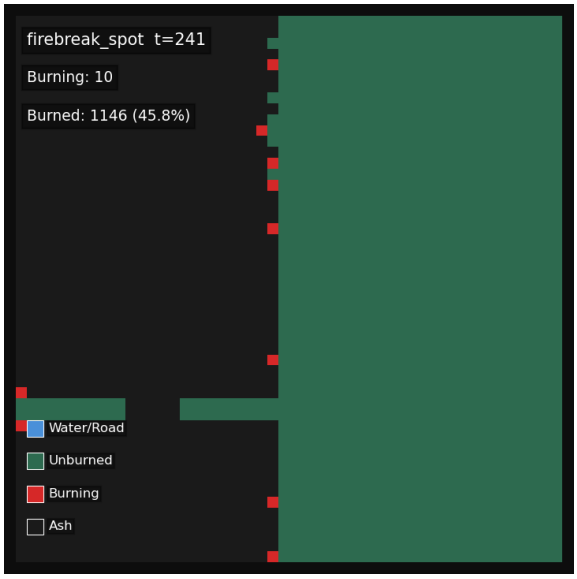

Fig. 4: Scenario 4 (Firebreaks + Spotting): isolated ignitions on the far bank; no sustained breach at 25 km/h.

G. Scenario 5 — Wildland–Urban Interface

This scenario places a road gap and an urban fuel patch (WUI) downstream of the ignition. Once the fire enters the WUI, intensity increases sharply due to the higher per-cell intensity of urban fuel.

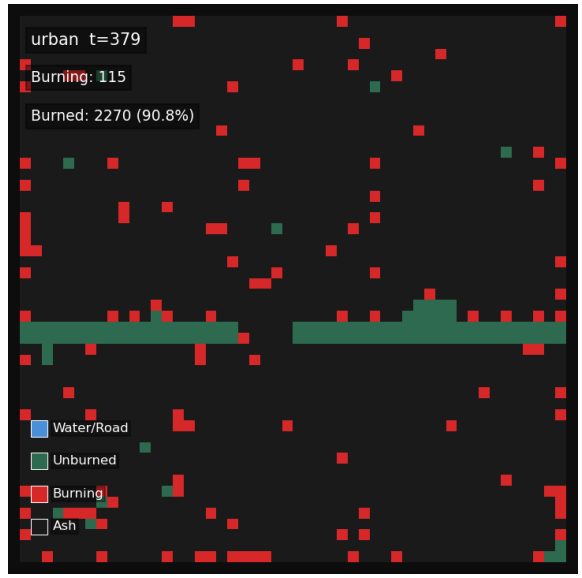

Fig. 5: Scenario 5 (Urban Interface): fire breaches the road gap and ignites urban structures; positive feedback accelerates urban spread.

H. Fort McMurray 2016 Validation

Scenarios 6 and 7 reproduce the May 3, 2016 conditions. The Athabasca River is encoded as a wide zero-weight barrier separating the forested ignition region from a mixed wildland/urban region.

1) Scenario 6 — Fort McMurray (No Spotting): With spotting disabled, the Athabasca River acts as a robust barrier. The fire is driven northeast by the strong southwest wind but halts at the river edge (Fig. 6).

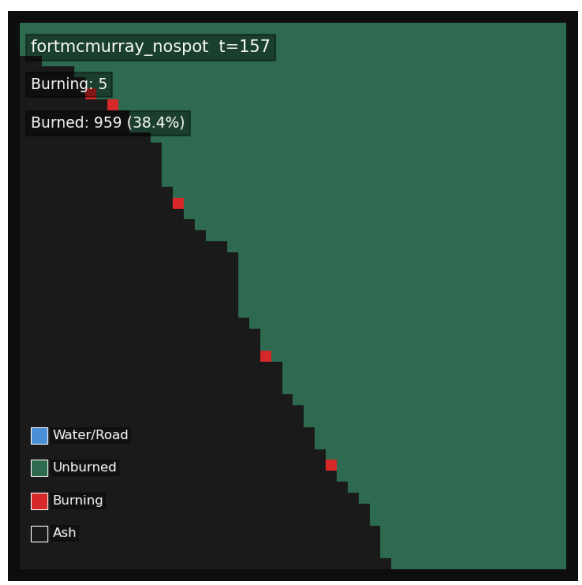

Fig. 6: Scenario 6 (Fort McMurray, no spotting): Athabasca River contains the fire under extreme wind when spotting is disabled.

2) Scenario 7 — Fort McMurray (With Spotting): With spotting enabled, embers cross the Athabasca River and ignite receptive fuels on the eastern bank. Secondary ignitions coalesce, and the fire establishes east of the river (Fig. 7).

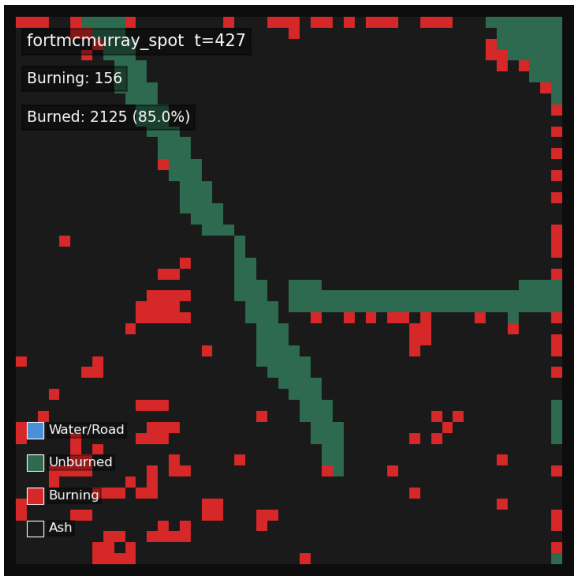

Fig. 7: Scenario 7 (Fort McMurray, with spotting): embers cross the river and ignite eastern fuels and urban areas.

I. Cross-Scenario Comparison

The seven scenarios collectively illustrate how directional physics, fuel heterogeneity, and spotting interact to shape wildfire behaviour. Several cross-cutting patterns emerge.

1) Wind as the Dominant Driver of Directionality: Scenarios 1 and 2 highlight the strong influence of wind. Moving from zero wind to 35 km/h transforms the fire's footprint from near-circular to highly elongated, with the long axis aligned to the wind vector. The asymmetric weights do all of the work: no conditional logic inside the transition function is required.

2) Firebreaks as Structural Constraints: Scenarios 3 and 4 show that rivers act as robust containment features in the absence of spotting, and as mostly-effective containment features under moderate spotting. The tiny 0.7 pp increase from Scenario 3 to Scenario 4 reinforces that isolated spot ignitions, once they occur on an unforested far bank, usually fail to sustain themselves.

3) Spotting as a Threshold-Driven Mechanism: Spotting becomes influential only when wind speed and ember production exceed critical thresholds. Below those thresholds, spot embers are too few, too short-ranged, or land in insufficiently receptive fuels. Above them, spotting can bypass arbitrary barriers. This regime change is the single largest qualitative contrast in the scenario set.

4) Urban Fuels Amplify Fire Intensity: Scenario 5 demonstrates rapid WUI escalation due to high-intensity urban fuels. The positive-feedback character of urban involvement is the mechanism behind the catastrophic urban damage observed in Scenario 7: once urban cells start burning, their long burn durations and high intensity sustain spread through the neighbourhood.

5) Extreme Conditions Produce Nonlinear Behaviour: The contrast between Scenarios 6 and 7 shows that spotting under extreme wind transforms a contained event into a catastrophic one. A linear sensitivity picture would massively understate the risk of such regime transitions; only scenario-based analysis reveals their existence.

J. Sensitivity Analysis

A qualitative sensitivity analysis was conducted on wind speed, fuel moisture, and spotting strength by sweeping each parameter across a representative range while holding the others at their Fort McMurray values.

Wind Speed Sensitivity:

- Below 10 km/h: spread is effectively symmetric; directional bias is dominated by the distance factor.
- 20–40 km/h: directional bias increases approximately linearly with wind speed; plumes stretch along the wind vector while lateral growth slows.
- Above 50 km/h: spotting dominates, and the burned area becomes highly sensitive to downwind fuel arrangement rather than local contiguity.

Fuel Moisture Sensitivity:

- High moisture suppresses ignition: a doubling of m_c roughly halves the per-step ignition probability via the $(1 - 0.85m_c)$ term.
- Low moisture amplifies both direct spread and spotting: dry fuels are more receptive to marginal embers, shifting the spotting regime threshold toward lower wind speeds.

Spotting Strength Sensitivity:

- Low *spot_base*: no sustained breaches regardless of wind; the model behaves like Scenario 6.
- Moderate *spot_base*: occasional breaches, often self-limiting; similar to Scenario 4.
- High *spot_base*: rapid downwind expansion; behaviour converges to Scenario 7.

These observations corroborate the qualitative threshold picture that emerges from the seven canonical scenarios.

K. Summary of Findings

The simulation results demonstrate that the Asymmetric Cell-DEVS model successfully captures:

- wind-driven anisotropy;
- fuel-driven variability in burn duration and intensity;
- structural containment by firebreaks;
- threshold-dependent spotting behaviour;
- rapid WUI escalation under urban fuel conditions;
- extreme-event dynamics consistent with the 2016 Fort McMurray wildfire.

These findings validate the model's ability to reproduce both routine and extreme wildfire behaviour across heterogeneous terrain.

V. Discussion

The scenario set reveals three recurring themes that generalise beyond any single configuration.

A. Directional Physics Belongs in the Topology

Across every non-symmetric scenario, the asymmetric weights did exactly what a Rothermel-style directional enhancement would do, but without any branching inside the transition function. The wind plume of Scenario 2, the containment of Scenario 3, and the spotting breach of Scenario 7 are all consequences of the same mechanism—precompiled per-edge weights—applied to different input data. This reinforces one of the core design claims of the paper: when directional physics is expressed structurally, the transition rule can stay simple without sacrificing realism.

B. Regime Changes Dominate Risk

The burned-area difference between Scenarios 6 and 7 is a factor of 2.2 for a single enabled mechanism. A purely parametric sensitivity sweep would not register this as a catastrophic risk because intermediate parameter values produce intermediate outcomes; it is only the combination of extreme wind, dry fuels, and non-zero spotting that flips the system into the breached regime. This is consistent with post-event analyses of real fires, which repeatedly identify "everything aligned" conditions as the trigger for catastrophic outcomes.

C. Urban Fuels are a Latent Amplifier

Urban cells do not change behaviour while the fire remains in the wildland, but they dominate the outcome once ignition reaches them. Scenario 5 shows this explicitly, and Scenario 7 shows the end state: 222 urban cells burned, matching the broad pattern of neighbourhood-scale loss documented at Fort McMurray. Any mitigation strategy that only considers wildland fuels will miss this amplification.

D. Validation against Fort McMurray

Quantitatively, the model reproduces the qualitative geography of the Horse River event: a southwest-driven plume, containment on the west bank when spotting is suppressed, and a successful east-bank breach with urban ignition when spotting is enabled at realistic strengths [11]. The match is qualitative rather than pixel-perfect—this is a 50×50 grid, not a plume-resolving CFD domain—but the behavioural regime is the one of interest, and it is recovered.

VI. Limitations

Several limitations are worth highlighting.

- **Static weather.** The scenarios use a single wind vector and FFMC value for each run. Real fires experience diurnal cycles, frontal passages, and plume-induced wind shifts. A dynamic weather layer would change the magnitude of spotting in Scenario 7, though the qualitative regime change would likely persist.
- **Surface fire only.** Crown fire transition, which is crucial in conifer-dominated landscapes, is represented only implicitly via fuel-type intensity parameters. There is no explicit canopy state, no ladder-fuel model, and no crown-to-surface feedback.
- **Simplified spotting kernel.** The $1/d^{1.5}$ decay and cone-shaped wind alignment are empirical shortcuts. Plume-coupled spotting would produce a richer landing distribution, especially in complex terrain where the plume itself is deflected.
- **No suppression.** The model does not represent aerial drops, retardant lines, firefighters, or controlled backburns. Real outcomes at Fort McMurray reflect heavy suppression effort that is absent from the simulation.
- **Grid resolution.** At 100 m per cell on a 50×50 grid, the domain covers only $25\,\mathrm{km}^2$. This is appropriate for illustrating mechanisms but coarse compared with operational domains.
- **Stochastic single-seed runs.** The tabulated scenarios use one random seed each. Ensemble averaging would give confidence intervals on burned area and would be a natural next step for quantitative validation.

None of these limitations undermines the core findings about directionality, firebreaks, and spotting; they instead mark the boundaries within which those findings are demonstrated.

VII. Future Work

Several extensions would increase both the realism and the utility of the framework.

- **Dynamic weather coupling.** Driving the topology's wind weights from a time-varying meteorological input (e.g., an hourly WRF trajectory) would let the model reproduce diurnal spread pulses and frontal shifts.
- **Crown fire module.** Adding a canopy state with its own ignition and burn-out rules would let the model distinguish surface from crown regimes, which matters for intensity and spotting ability.
- **Suppression models.** Introducing suppression nodes (aerial drops, ground crews, retardant lines) as dynamic zero-weight edges would let the model study mitigation strategies and their timing.
- **ML-based parameter tuning.** Surrogate models trained on ensemble outputs could invert observed burned-area patterns to identify likely values of *spot_base* and other unknown parameters for historical fires.
- **Scaling to operational domains.** Coupling the Cadmium implementation with tiled topology generation would enable simulations at the scale of full watersheds or municipalities.

- **Interactive visualization.** A real-time dashboard, potentially built on top of the existing Python visualization suite, would make the framework useful for scenario planning and public communication.

VIII. Conclusions

This project developed and evaluated an Asymmetric Cell-DEVS wildfire model capable of reproducing a wide range of fire behaviours across heterogeneous landscapes. The model was motivated by the need for computationally efficient, spatially explicit wildfire simulations that capture directional effects such as wind-driven spread, fuel heterogeneity, firebreak interactions, and spotting. By integrating asymmetric vicinity weights, fuel-dependent ignition rules, and a probabilistic spotting mechanism, the model provides a flexible framework for studying both routine and extreme wildfire dynamics.

The simulation results demonstrate that the model successfully reproduces several well-established wildfire phenomena. Under calm conditions, the system exhibits symmetric radial spread, confirming that the underlying transition rules do not introduce artificial bias. Wind-driven scenarios produce elongated plumes and strong directional anisotropy, while structural firebreaks such as rivers act as robust containment features in the absence of spotting. The model also captures threshold-driven transitions: moderate spotting produces isolated ignitions without sustained breaches, whereas extreme wind and high ember production enable long-range spotting and river crossing, consistent with documented behaviour from the 2016 Fort McMurray wildfire [11]. Urban-interface simulations further show rapid intensity escalation once the fire enters high-energy built fuels, reflecting real-world WUI vulnerabilities [12].

Methodologically, the work supports a broader claim about modelling style: when physical directionality is embedded in the topology rather than encoded as special cases inside the transition function, the resulting model is simpler, faster, and easier to audit. The Fort McMurray scenarios, in particular, show that the qualitative regime change associated with a catastrophic event can be reproduced by flipping a small number of topology parameters, which is exactly what a mitigation-oriented user of the model would want.

Overall, the Asymmetric Cell-DEVS formulation provides a credible and computationally efficient approach for modelling wildfire spread across complex terrain. The model's ability to reproduce both controlled and extreme scenarios highlights its potential for exploratory analysis, scenario testing, and educational use. While the framework captures key mechanisms such as wind forcing, fuel variation, and spotting, future work can incorporate additional physical processes—such as dynamic weather, crown-fire transition, or suppression strategies—to further enhance realism and predictive capability. The findings of this study reinforce the value of discrete-event, cellular modelling approaches for understanding wildfire behaviour and evaluating the influence of environmental and structural factors on fire propagation.

References

[1] J. M. Morales and G. Wainer, "Design and implementation of a Cell-DEVS model for fire spreading," in *Proc. 2003 Winter Simulation Conference*, 2003, pp. 930–938.

[2] G. Wainer, *Discrete-Event Modeling and Simulation: A Practitioner's Approach*. Boca Raton, FL, USA: CRC Press, 2009.

[3] G. Wainer and N. Giambiasi, "N-dimensional Cell-DEVS models," *Discrete Event Dynamic Systems*, vol. 14, no. 1, pp. 47–76, 2004.

[4] A. L. Sullivan, "Wildland surface fire spread modelling, 1990–2007. 1: Physical and quasi-physical models," *International Journal of Wildland Fire*, vol. 18, no. 4, pp. 349–368, 2009.

[5] A. L. Sullivan, "Wildland surface fire spread modelling, 1990–2007. 2: Empirical and quasi-empirical models," *International Journal of Wildland Fire*, vol. 18, no. 4, pp. 369–386, 2009.

[6] M. A. Finney, "FARSITE: Fire area simulator—Model development and evaluation," USDA Forest Service, RMRS-RP-4, 1998.

[7] H. E. Anderson, "Aids to determining fuel models for estimating fire behavior," USDA Forest Service, INT-GTR-122, 1982.

[8] F. A. Albini, "Spot fire distance from burning trees—A predictive model," USDA Forest Service, INT-RP-56, 1979.

[9] S. W. Taylor and M. E. Alexander, "Science, technology, and human factors in wildfire management," *Current Forestry Reports*, vol. 2, pp. 1–14, 2016.

[10] J. L. Coen et al., "WRF-Fire: Coupled weather–wildland fire modeling with the Weather Research and Forecasting model," *Journal of Applied Meteorology and Climatology*, vol. 52, no. 1, pp. 16–38, 2013.

[11] Natural Resources Canada, "Fort McMurray 2016: A wildfire review," Canadian Forest Service, Tech. Rep., 2017.

[12] L. M. Johnston and S. W. Flannigan, "Mapping Canadian wildland fire interface areas," *International Journal of Wildland Fire*, vol. 24, no. 1, pp. 1–14, 2015.

[13] R. C. Rothermel, "A mathematical model for predicting fire spread in wildland fuels," USDA Forest Service, INT-RP-115, 1972.

[14] D. Morvan, "Physical phenomena and length scales governing the behaviour of wildfires: A case for physical modelling," *Fire Technology*, vol. 47, pp. 437–460, 2011.

[15] E. Pastor, L. Zárate, E. Planas, and J. Arnaldos, "Mathematical models and calculation systems for the study of wildland fire behaviour," *Progress in Energy and Combustion Science*, vol. 29, no. 2, pp. 139–153, 2003.